# Supplementary Material

## (NAMAT: Node-wise Message-Aware Multi-Graph Attention for Cancer Driver Gene Prioritization)

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## 1 Supplementary Methods

### 1.1 Dataset construction and label definition

Twelve cancer-specific benchmark datasets were utilized, adhering to the experimental settings established by EMOGI[1] and MTGCN[2]. For each cancer type, positive samples were defined as known driver genes annotated in the Network of Cancer Genes (NCG)[3]. Negative samples were defined as putative non-driver genes obtained from the MTGCN benchmark after excluding genes associated with established cancer-related resources, including NCG, COSMIC[4], OMIM[5], and KEGG cancer pathways[6]. This filtering reduces the inclusion of known cancer-associated genes among negatives but does not establish experimentally confirmed non-driver status.

A consistent set of 2,187 negative genes was applied across all cancer types. The number of positive driver genes varied by cancer type, reflecting differences in available driver-gene annotations. Dataset statistics are provided in Supplementary Table S1.

**Table S1.** Dataset statistics across 12 cancer types. Positive samples correspond to driver genes, negative samples correspond to putative non-driver genes.

| | BRCA | BLCA | LUAD | THCA | LIHC | LUSC | ESCA | PRAD | STAD | COAD | UCEC | CESC |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Driver genes | 202 | 95 | 179 | 64 | 82 | 26 | 71 | 50 | 73 | 141 | 68 | 17 |
| Negative genes | 2187 | 2187 | 2187 | 2187 | 2187 | 2187 | 2187 | 2187 | 2187 | 2187 | 2187 | 2187 |

### 1.2 PPI networks

We used the same six PPI networks as the EMOGI benchmark: CPDB[7], STRING-db[8], MultiNet[9], PCNet[10], IRefIndex, and IRefIndex_2015[11].

### 1.3 Implementation details and hyperparameters

NAMAT was implemented using PyTorch. The input feature encoder consisted of a two- or three-layer multilayer perceptron with hidden dimensions of 64 and an output dimension of 64. The graph backbone incorporated two message-passing layers with hidden dimensions of 64 and 128. Dropout with a probability of 0.3 or 0.4 was applied to the input encoder, message-passing layers, and prediction head. Additionally, feature-wise dropout with a probability of 0.2 was applied to the input features during training. Prior to model input, node features were standardized using column-wise z-score normalization. The input MLP depth, dropout probability, and learning rate were selected by grid search, as described at the end of this section.

The attention MLP in each message-passing layer contains two linear layers with an intermediate ReLU activation and a scalar output. Its input concatenates the current node representation, graph-specific message, and log-degree, giving architectures of $129 \to 64 \to 1$ and $193 \to 128 \to 1$ for the first and second layers, respectively. Each layer uses a separate attention MLP shared across all six PPI networks, with no internal dropout.

The model was optimized using the AdamW optimizer with weight decay. The base learning rate was set to $10^{-4}$ or $10^{-3}$, employing a linear warm-up for 5 epochs followed by cosine decay. Weight decay was set to $4 \times 10^{-4}$. The training objective utilized weighted binary cross-entropy with logits, a label smoothing coefficient of $\varepsilon = 0.06$, and a positive-class weight $w = \min(N_{\text{neg}}/N_{\text{pos}}, 10)$ to address class imbalance.

Gradient clipping used a maximum norm of 1.0. Training ran for at most 380 epochs, with early-stopping patience of 80 based on validation loss. The lowest-validation-loss checkpoint was retained. Attention warm-up, PPI dropout, and the temperature schedule are detailed in Section 1.4.

The primary predictive-performance experiments used five-fold stratified cross-validation repeated over 10 runs with different random seeds. Test AUPRC was averaged over the five folds within each run, and performance was summarized as the mean and standard deviation across the 10 run-level values. The temperature-sensitivity experiments and computational-cost benchmark used the separate protocols described below.

Hyperparameters were selected by grid search within each cross-validation fold over dropout $\{0.3, 0.4\}$, learning rate $\{10^{-4}, 10^{-3}\}$, and input MLP depth $\{2, 3\}$, with the temperature schedule held fixed. Within each outer fold, each of the eight configurations was trained on the training partition and evaluated on the validation partition. The configuration with the highest validation AUPRC was retrained on the training partition and evaluated once on the test fold, so test labels were never used for selection. This procedure was applied to all reported NAMAT results except the temperature-sensitivity analysis and the computational-cost benchmark, which used the default configuration (dropout 0.4, learning rate $10^{-3}$, and a three-layer input MLP). The code supports both the grid-search mode and this default configuration.

### 1.4 Temperature schedule and sensitivity analysis

Attention weights are fixed uniformly over valid PPI networks for the first 30 epochs. Subsequently, learned weighting and PPI dropout ($p = 0.25$) are enabled. The training temperature follows

$$\tau_e = 0.6 + 0.2\frac{379 - e}{379}, \qquad e = 0, \ldots, 379. \tag{1}$$

The schedule is shared across cancer types. Temperature first affects learned weighting at epoch 31 ($\tau_e \approx 0.784$), and early stopping may terminate training before it reaches 0.6. Validation, checkpoint selection, and inference always use $\tau = 0.6$, with PPI dropout disabled.

Sensitivity experiments on BRCA, BLCA, and STAD retained the manuscript splits and all other settings, using the default configuration (dropout 0.4, learning rate $10^{-3}$, and a three-layer input MLP) without grid search. The fixed-temperature sweep used one run with five folds per cancer; the annealing comparison used three runs with five folds each. Validation and inference used each configuration's fixed temperature or schedule endpoint. Seeds were reset per fold in the sweep and per run in the annealing comparison; results are therefore compared within each experiment. These are sensitivity analyses, not selection based on test performance.

Table S2. Fixed-temperature sensitivity: test AUPRC averaged over five folds from one run. Average summarizes the three cancers. Ref. is the manuscript schedule $0.8 \to 0.6$. All configurations retain the 30-epoch uniform-weight warm-up.

| Cancer | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 | 1.0 | 2 | 4 | 6 | 10 | Ref. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **BRCA** | 0.804 | 0.807 | 0.805 | 0.804 | 0.803 | 0.801 | 0.801 | 0.800 | 0.799 | 0.799 | 0.794 | 0.796 | 0.792 | 0.785 | 0.803 |
| **BLCA** | 0.794 | 0.789 | 0.789 | 0.793 | 0.795 | 0.794 | 0.793 | 0.794 | 0.794 | 0.794 | 0.789 | 0.784 | 0.783 | 0.781 | 0.792 |
| **STAD** | 0.790 | 0.786 | 0.788 | 0.785 | 0.785 | 0.786 | 0.786 | 0.786 | 0.783 | 0.782 | 0.780 | 0.778 | 0.772 | 0.771 | 0.787 |
| **Average** | 0.796 | 0.794 | 0.794 | 0.794 | 0.794 | 0.794 | 0.793 | 0.793 | 0.792 | 0.792 | 0.788 | 0.786 | 0.782 | 0.779 | 0.794 |

Table S3. Fixed versus annealed temperature: test AUPRC averaged over three runs, each comprising five folds per cancer. Average and SD summarize the three run-level averages across cancers. The manuscript schedule is $0.8 \to 0.6$.

| | Fixed temperature | | Linear annealing | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Cancer** | 0.1 | 0.6 | $0.9 \to 0.7$ | $0.8 \to 0.6$ | $0.7 \to 0.5$ | $0.5 \to 0.3$ | $0.3 \to 0.1$ | $0.5 \to 0.1$ | $1.0 \to 0.1$ |
| **BRCA** | 0.800 | 0.803 | 0.803 | 0.803 | 0.801 | 0.802 | 0.801 | 0.798 | 0.798 |
| **BLCA** | 0.791 | 0.791 | 0.792 | 0.792 | 0.791 | 0.789 | 0.786 | 0.784 | 0.769 |
| **STAD** | 0.788 | 0.778 | 0.778 | 0.778 | 0.777 | 0.782 | 0.772 | 0.773 | 0.758 |
| **Average** | 0.793 | 0.791 | 0.791 | 0.791 | 0.790 | 0.791 | 0.786 | 0.785 | 0.775 |
| **SD** | 0.003 | 0.005 | 0.007 | 0.007 | 0.006 | 0.005 | 0.006 | 0.006 | 0.004 |

NAMAT was relatively insensitive to fixed temperatures between 0.1 and 1.0, with average AUPRCs of 0.792–0.796; larger tested temperatures yielded lower averages (Table S2). In the repeated comparison, the manuscript schedule and fixed $\tau = 0.6$

both achieved an average AUPRC of 0.791 at three-decimal precision. Its average differed by at most approximately 0.0015 from schedules $0.9 \rightarrow 0.7$, $0.7 \rightarrow 0.5$, and $0.5 \rightarrow 0.3$, whereas $1.0 \rightarrow 0.1$ yielded 0.775 (Table S3). These findings support retaining $0.8 \rightarrow 0.6$ without claiming uniquely optimal endpoints or a consistent advantage over fixed temperature.

### 1.5 Baseline comparison protocol

NAMAT was compared with EMOGI[1], MTGCN[2], MRNGCN[12], DGHNN[13], MODIG[14], and EMGNN[15]. All reimplemented methods used the same cancer-specific benchmark labels, gene-level splits, and evaluation metrics, as supported by each method. AUPRC assessed classification performance, while Hits@K assessed driver recovery using a consistent ranking candidate pool. EMOGI, MTGCN, DGHNN, MODIG, and EMGNN were reimplemented in PyTorch and run with the best hyperparameter settings reported in their original publications, without additional tuning on our benchmark. MRNGCN AUPRC values were taken from its original publication, which used the same benchmark and evaluation protocol but not the identical gene-level splits. Because per-gene predictions were unavailable, MRNGCN was excluded from the Hits@K analysis.

### 1.6 Derivation of the −MAA ablation

The −MAA ablation removes graph-specific information from the attention input while retaining multi-graph message passing. It tests adaptive node-wise weighting against uniform averaging.

In the full NAMAT model, the graph-specific evidence descriptor is defined as

$$\phi_{i,k}^{(t)} = \left[ m_{i,k}^{(t)} \,\|\, \log(1+\deg_k(i)) \right], \tag{2}$$

and the attention input is

$$u_{i,k}^{(t)} = \left[ h_i^{(t)} \,\|\, \phi_{i,k}^{(t)} \right]. \tag{3}$$

The attention score is then computed as

$$s_{i,k}^{(t)} = g^{(t)}\left( u_{i,k}^{(t)} \right). \tag{4}$$

Because $u_{i,k}^{(t)}$ contains graph-specific information, different PPI networks can receive different attention scores for the same node.

In the −MAA setting, the attention input is restricted to the current node state only:

$$u_{i,k}^{(t)} = h_i^{(t)}, \quad \forall k. \tag{5}$$

For a fixed node $i$ and layer $t$, $h_i^{(t)}$ is identical across all valid graphs. Therefore, all valid graphs receive the same attention score:

$$s_{i,k}^{(t)} = g^{(t)}\left( h_i^{(t)} \right), \quad \forall k. \tag{6}$$

Applying the softmax over identical scores gives a uniform distribution over valid PPI networks:

$$\alpha_{i,k}^{(t)} = \frac{1}{K_i}, \tag{7}$$

where $K_i$ denotes the number of valid PPI networks for node $i$. Let $\mathcal{K}_i = \{k : \deg_k(i) > 0\}$ denote their index set, so that $K_i = |\mathcal{K}_i|$. Thus, the aggregated message becomes

$$\tilde{m}_i^{(t)} = \frac{1}{K_i} \sum_{k \in \mathcal{K}_i} m_{i,k}^{(t)}. \tag{8}$$

Thus, −MAA retains graph-specific messages but weights valid networks equally. Its comparison with NAMAT evaluates the contribution of message-aware adaptive weighting.

### 1.7 Ablation experiment configuration

Controlled comparisons evaluated message-aware attention, the input MLP, residual connections, and graph-specific messages. The −MAA variant retains message passing but uniformly averages valid graph-specific messages, as derived above. The −MLP variant removes the nonlinear input encoder, while −Res removes the residual pathway, leaving the aggregated graph message as the input to the subsequent normalization and activation. The −MSG variant removes graph-specific messages from the node update while retaining the input MLP and residual pathway; it evaluates graph-derived information rather than the fusion mechanism.

All ablation variants were evaluated using the same datasets, train/validation/test splits, optimization protocol, and evaluation metrics as the full NAMAT model.

## 2 Supplementary Results

### 2.1 Detailed AUPRC performance

**Table S4.** Detailed AUPRC comparison for 12 cancer types. MTG, DGH, and MRNG denote MTGCN, DGHNN, and MRNGCN, respectively. Results are reported as mean AUPRC ± standard deviation over repeated runs, rounded to three decimals. MRNGCN values are taken from its original publication. The top result for each cancer type is shown in **bold**. A superscript * indicates that NAMAT significantly outperformed EMGNN according to the Wilcoxon signed-rank test ($p < 0.05$).

| Cancer | NAMAT | EMOGI | MTG | DGH | MRNG | MODIG | EMGNN |
|---|---|---|---|---|---|---|---|
| **BRCA** | **0.802**±0.006* | 0.648±0.005 | 0.658±0.004 | 0.683±0.006 | 0.692±0.004 | 0.586±0.005 | 0.766±0.008 |
| **BLCA** | **0.789**±0.005* | 0.549±0.010 | 0.657±0.008 | 0.694±0.006 | 0.725±0.005 | 0.681±0.025 | 0.752±0.018 |
| **LUAD** | **0.744**±0.008* | 0.559±0.011 | 0.628±0.010 | 0.641±0.009 | 0.629±0.012 | 0.603±0.003 | 0.710±0.009 |
| **LIHC** | **0.595**±0.019* | 0.385±0.019 | 0.465±0.020 | 0.481±0.019 | 0.547±0.015 | 0.253±0.025 | 0.546±0.021 |
| **THCA** | **0.426**±0.026* | 0.259±0.008 | 0.291±0.011 | 0.315±0.023 | 0.308±0.010 | 0.028±0.001 | 0.362±0.032 |
| **LUSC** | **0.616**±0.053* | 0.222±0.027 | 0.310±0.041 | 0.405±0.054 | 0.348±0.032 | 0.071±0.014 | 0.524±0.048 |
| **ESCA** | **0.611**±0.015* | 0.417±0.030 | 0.472±0.028 | 0.476±0.015 | 0.537±0.012 | 0.264±0.019 | 0.568±0.023 |
| **PRAD** | **0.827**±0.009* | 0.523±0.022 | 0.614±0.019 | 0.706±0.013 | 0.669±0.016 | 0.383±0.097 | 0.783±0.035 |
| **STAD** | **0.779**±0.009* | 0.460±0.015 | 0.593±0.012 | 0.645±0.008 | 0.646±0.011 | 0.552±0.051 | 0.743±0.014 |
| **COAD** | **0.610**±0.008* | 0.346±0.007 | 0.382±0.009 | 0.447±0.069 | 0.495±0.007 | 0.443±0.066 | 0.587±0.006 |
| **UCEC** | **0.743**±0.014 | 0.400±0.014 | 0.495±0.015 | 0.534±0.025 | 0.570±0.015 | 0.389±0.080 | 0.726±0.025 |
| **CESC** | **0.655**±0.030 | 0.555±0.052 | 0.597±0.053 | 0.564±0.041 | 0.534±0.069 | 0.059±0.046 | 0.640±0.042 |

Table S4 reports mean AUPRC and standard deviation across repeated runs. NAMAT achieved the highest mean AUPRC in all 12 cancer types. Compared with EMGNN, the strongest baseline on average, NAMAT showed significant improvements in 10 cancer types (Wilcoxon signed-rank test, $p < 0.05$, marked by *). Improvements in UCEC and CESC were not statistically significant.

### 2.2 Comprehensive Hits@K performance analysis

Table S5 reports Hits@K for $K \in \{1, 3, 5, 10, 20, 50\}$ across 12 cancer types. NAMAT achieved the highest average recovery at every evaluated cutoff. MRNGCN was excluded because its results are taken from the original publication, which does not provide per-gene predictions.

**Table S5.** Comparison of Hits@K across 12 cancer types for K=1,3,5,10,20,50. Higher values correspond to greater recovery of known driver genes within the top-K ranked candidates. The best result for each cancer type and cutoff is highlighted in **bold**.

| K | Method | BRCA | BLCA | LUAD | LIHC | THCA | LUSC | ESCA | PRAD | STAD | COAD | UCEC | CESC | Avg |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | **NAMAT** | **0.46** | **0.36** | 0.28 | **0.26** | **0.40** | **0.20** | **0.86** | **0.28** | 0.28 | **0.94** | 0.44 | **0.20** | **0.41** |
| | **EMOGI** | 0.26 | 0.16 | 0.06 | 0.08 | 0.00 | 0.10 | 0.04 | 0.12 | 0.06 | 0.16 | 0.06 | 0.00 | 0.09 |
| | **MTGCN** | 0.20 | 0.16 | 0.20 | 0.08 | 0.04 | 0.02 | 0.02 | 0.04 | 0.04 | 0.02 | 0.06 | 0.00 | 0.07 |
| | **DGHNN** | 0.20 | 0.00 | 0.20 | 0.00 | 0.00 | 0.00 | 0.20 | 0.20 | 0.20 | 0.00 | 0.40 | 0.00 | 0.12 |
| | **MODIG** | 0.32 | 0.30 | 0.28 | 0.12 | 0.00 | 0.00 | 0.40 | 0.12 | 0.14 | 0.40 | 0.38 | 0.00 | 0.21 |
| | **EMGNN** | 0.40 | 0.26 | **0.34** | 0.20 | 0.36 | 0.18 | 0.58 | 0.26 | **0.34** | 0.86 | **0.52** | 0.18 | 0.37 |
| 3 | **NAMAT** | **1.08** | **1.74** | **0.72** | **0.80** | **0.68** | 0.36 | **1.60** | 0.82 | **1.26** | **2.14** | **1.22** | **0.26** | **1.06** |
| | **EMOGI** | 0.60 | 0.34 | 0.34 | 0.20 | 0.00 | 0.10 | 0.14 | 0.22 | 0.30 | 0.34 | 0.38 | 0.00 | 0.25 |
| | **MTGCN** | 0.38 | 0.24 | 0.26 | 0.16 | 0.08 | 0.02 | 0.18 | 0.08 | 0.22 | 0.18 | 0.12 | 0.06 | 0.17 |
| | **DGHNN** | 0.60 | 0.40 | 0.20 | 0.00 | 0.00 | **0.40** | 0.20 | 0.40 | 0.20 | 0.00 | 0.40 | 0.00 | 0.23 |
| | **MODIG** | 0.92 | 1.00 | 0.56 | 0.20 | 0.00 | 0.00 | 0.46 | 0.26 | 0.40 | 1.12 | 0.60 | 0.00 | 0.46 |
| | **EMGNN** | **1.08** | 1.40 | 0.66 | 0.78 | 0.66 | 0.30 | 1.18 | **0.96** | 1.14 | 1.80 | 1.06 | 0.20 | 0.94 |
| 5 | **NAMAT** | 1.86 | **2.70** | **1.00** | **0.96** | **0.80** | 0.44 | **2.08** | 1.06 | **1.46** | **2.58** | **1.68** | 0.68 | **1.44** |
| | **EMOGI** | 0.96 | 0.42 | 0.38 | 0.30 | 0.00 | 0.12 | 0.14 | 0.32 | 0.64 | 0.42 | 0.48 | 0.00 | 0.35 |
| | **MTGCN** | 0.80 | 0.36 | 0.50 | 0.22 | 0.08 | 0.04 | 0.26 | 0.18 | 0.50 | 0.40 | 0.26 | 0.14 | 0.31 |
| | **DGHNN** | 1.00 | 0.80 | 0.20 | 0.00 | 0.20 | **0.60** | 0.40 | 0.80 | 0.40 | 0.20 | 0.80 | 0.00 | 0.45 |
| | **MODIG** | 0.92 | 1.20 | 0.68 | 0.26 | 0.00 | 0.00 | 0.54 | 0.52 | 0.72 | 1.46 | 0.64 | 0.00 | 0.58 |
| | **EMGNN** | **1.92** | 2.28 | 0.96 | 0.92 | 0.68 | 0.36 | 1.30 | **1.10** | 1.36 | 2.14 | 1.60 | **0.82** | 1.29 |
| 10 | **NAMAT** | **3.66** | **4.44** | **1.48** | 1.18 | **1.04** | 0.56 | **2.72** | **2.12** | **2.14** | **3.22** | **2.76** | **1.10** | **2.20** |
| | **EMOGI** | 1.44 | 0.56 | 0.78 | 0.36 | 0.20 | 0.14 | 0.32 | 0.44 | 0.76 | 0.56 | 0.74 | 0.02 | 0.53 |
| | **MTGCN** | 1.66 | 0.56 | 0.84 | 0.24 | 0.16 | 0.08 | 0.30 | 0.36 | 0.64 | 0.58 | 0.50 | 0.24 | 0.51 |
| | **DGHNN** | 1.60 | 1.40 | 0.40 | 0.20 | 0.20 | **0.60** | 0.60 | 1.80 | 1.00 | 0.20 | 1.40 | 0.40 | 0.82 |
| | **MODIG** | 1.52 | 1.80 | 0.96 | 0.46 | 0.00 | 0.00 | 0.72 | 0.92 | 1.26 | 1.66 | 0.88 | 0.00 | 0.85 |
| | **EMGNN** | 3.44 | 3.98 | 1.42 | **1.28** | 0.84 | 0.50 | 1.78 | 1.48 | 1.90 | 2.82 | 2.40 | 1.06 | 1.91 |
| 20 | **NAMAT** | **4.52** | **5.90** | **2.28** | 1.72 | **1.26** | 0.70 | **3.36** | **2.70** | **3.24** | **4.12** | **3.38** | **1.20** | **2.87** |
| | **EMOGI** | 1.82 | 0.62 | 1.04 | 0.46 | 0.26 | 0.16 | 0.40 | 0.52 | 0.86 | 0.80 | 1.04 | 0.04 | 0.67 |
| | **MTGCN** | 2.20 | 0.72 | 1.34 | 0.24 | 0.22 | 0.12 | 0.56 | 0.82 | 0.76 | 0.72 | 0.76 | 0.34 | 0.73 |
| | **DGHNN** | 2.60 | 2.00 | 1.40 | 0.40 | 0.20 | 0.60 | 0.80 | 2.60 | 1.80 | 0.20 | 1.80 | 0.40 | 1.23 |
| | **MODIG** | 2.40 | 3.40 | 1.46 | 0.60 | 0.00 | 0.00 | 0.86 | 1.00 | 1.86 | 2.12 | 1.02 | 0.08 | 1.23 |
| | **EMGNN** | 4.40 | 5.10 | 1.98 | **1.82** | 1.22 | **0.72** | 2.62 | 2.58 | 2.90 | 3.84 | 2.94 | 1.18 | 2.61 |
| 50 | **NAMAT** | **8.10** | **7.30** | **4.78** | **3.52** | **1.60** | **1.00** | **3.94** | **4.26** | **4.82** | **5.38** | **4.76** | 1.34 | **4.23** |
| | **EMOGI** | 3.06 | 1.34 | 2.04 | 0.76 | 0.44 | 0.20 | 0.92 | 1.08 | 1.36 | 1.54 | 1.40 | 0.34 | 1.21 |
| | **MTGCN** | 3.20 | 1.54 | 2.22 | 0.40 | 0.26 | 0.22 | 0.96 | 1.26 | 1.46 | 1.14 | 1.08 | 0.48 | 1.19 |
| | **DGHNN** | 4.40 | 3.20 | 3.60 | 1.40 | 0.44 | 0.60 | 1.00 | 3.80 | 3.22 | 0.60 | 2.80 | 0.80 | 2.15 |
| | **MODIG** | 3.96 | 5.80 | 3.46 | 1.00 | 0.00 | 0.08 | 1.46 | 1.72 | 3.26 | 3.32 | 1.70 | 0.12 | 2.16 |
| | **EMGNN** | 7.08 | 7.10 | 4.22 | 3.30 | 1.46 | 0.84 | 3.74 | 3.92 | 5.00 | 5.12 | 4.22 | **1.36** | 3.95 |

### 2.3 Individual-PPI network performance

Table S6 compares NAMAT with six single-PPI variants. NAMAT achieved higher mean AUPRC than every individual-PPI variant across all 12 cancer types. PCNet was the strongest single network on average (0.635), compared with 0.683 for NAMAT, supporting the benefit of combining complementary interaction sources.

**Table S6.** AUPRC comparison between NAMAT and models trained on individual PPI networks across 12 cancer types. Results are reported as mean AUPRC ± standard deviation over repeated runs, rounded to three decimals. IRef and IRef2015 denote IRefIndex and IRefIndex_2015, respectively. The best result for each cancer type is shown in **bold**.

| Cancer | NAMAT | CPDB | STRING | MultiNet | PCNet | IRef | IRef2015 |
|---|---|---|---|---|---|---|---|
| **BRCA** | **0.802**±0.006 | 0.705±0.008 | 0.704±0.006 | 0.720±0.006 | 0.750±0.009 | 0.686±0.008 | 0.717±0.009 |
| **BLCA** | **0.789**±0.005 | 0.743±0.010 | 0.727±0.008 | 0.727±0.007 | 0.750±0.005 | 0.705±0.010 | 0.740±0.008 |
| **LUAD** | **0.744**±0.008 | 0.681±0.008 | 0.651±0.008 | 0.634±0.013 | 0.706±0.007 | 0.664±0.006 | 0.663±0.008 |
| **LIHC** | **0.595**±0.019 | 0.538±0.009 | 0.532±0.015 | 0.524±0.017 | 0.576±0.017 | 0.488±0.009 | 0.509±0.015 |
| **THCA** | **0.426**±0.026 | 0.341±0.015 | 0.299±0.016 | 0.305±0.015 | 0.351±0.024 | 0.291±0.024 | 0.345±0.017 |
| **LUSC** | **0.616**±0.053 | 0.413±0.020 | 0.420±0.041 | 0.411±0.047 | 0.539±0.031 | 0.475±0.040 | 0.361±0.048 |
| **ESCA** | **0.611**±0.015 | 0.535±0.016 | 0.522±0.010 | 0.498±0.015 | 0.557±0.018 | 0.535±0.011 | 0.530±0.011 |
| **PRAD** | **0.827**±0.009 | 0.723±0.019 | 0.755±0.014 | 0.764±0.035 | 0.744±0.009 | 0.742±0.020 | 0.754±0.017 |
| **STAD** | **0.779**±0.009 | 0.718±0.012 | 0.713±0.006 | 0.700±0.012 | 0.759±0.010 | 0.728±0.014 | 0.723±0.011 |
| **COAD** | **0.610**±0.008 | 0.575±0.007 | 0.566±0.010 | 0.553±0.007 | 0.588±0.005 | 0.567±0.009 | 0.570±0.009 |
| **UCEC** | **0.743**±0.014 | 0.710±0.015 | 0.707±0.011 | 0.723±0.011 | 0.736±0.012 | 0.726±0.015 | 0.733±0.017 |
| **CESC** | **0.655**±0.030 | 0.534±0.052 | 0.541±0.042 | 0.577±0.050 | 0.565±0.043 | 0.505±0.036 | 0.599±0.021 |

**Incremental contribution of individual PPI networks.** A greedy analysis on BLCA started with PCNet, the strongest single-PPI variant. At each step, each remaining network was added separately, and the combination with the highest AUPRC was retained.

Table S7 shows that adding MultiNet increased AUPRC from 0.7500 to 0.7679 ($p = 7.02 \times 10^{-8}$). Subsequent inclusion of CPDB and STRING increased AUPRC to 0.7893. Adding IRefIndex or IRefIndex_2015 provided no significant further improvement. These results indicate diminishing predictive gains from additional networks in this BLCA analysis.

**Table S7.** Incremental contribution of individual PPI networks to NAMAT performance on BLCA. Network indices denote individual PPI networks: 1 = CPDB, 2 = STRING, 3 = MultiNet, 4 = PCNet, 5 = IRefIndex, and 6 = IRefIndex2015.

| Baseline configuration | Added network | AUPRC after | Mean diff. | $p$-value |
|---|---|---|---|---|
| 4 (0.7500±0.0054) | +1 (CPDB) | 0.7614±0.0071 | +0.0112 | 0.0007 |
| | +2 (STRING) | 0.7658±0.0069 | +0.0156 | $5.82 \times 10^{-6}$ |
| | +3 (MultiNet) | 0.7679±0.0070 | +0.0177 | $7.02 \times 10^{-8}$ |
| | +5 (IRefIndex) | 0.7604±0.0077 | +0.0102 | $4.96 \times 10^{-5}$ |
| | +6 (IRefIndex2015) | 0.7606±0.0075 | +0.0104 | $4.73 \times 10^{-5}$ |
| 4+3 (0.7679±0.0070) | +1 (CPDB) | 0.7818±0.0072 | +0.0139 | $8.45 \times 10^{-6}$ |
| | +2 (STRING) | 0.7791±0.0071 | +0.0112 | $2.39 \times 10^{-6}$ |
| | +5 (IRefIndex) | 0.7798±0.0055 | +0.0119 | $1.26 \times 10^{-6}$ |
| | +6 (IRefIndex2015) | 0.7808±0.0051 | +0.0129 | $4.44 \times 10^{-6}$ |
| 4+3+1 (0.7818±0.0072) | +2 (STRING) | 0.7893±0.0066 | +0.0075 | 0.0001 |
| | +5 (IRefIndex) | 0.7835±0.0057 | +0.0017 | 0.0495 |
| | +6 (IRefIndex2015) | 0.7839±0.0059 | +0.0021 | 0.0242 |
| 4+3+1+2 (0.7893±0.0066) | +5 (IRefIndex) | 0.7891±0.0070 | -0.0002 | 0.3616 |
| | +6 (IRefIndex2015) | 0.7876±0.0055 | -0.0017 | 0.2188 |
| 4+3+1+2+5 (0.7891±0.0070) | +6 (IRefIndex2015) | 0.7890±0.0050 | -0.0001 | 0.2680 |

### 2.4 Detailed ablation results

Supplementary Table S8 presents the complete per-cancer ablation results. Each block compares two matched configurations that differ only in the component under investigation, enabling a controlled assessment of the contributions from message-aware attention, input feature transformation, residual connections, and graph-based message passing. The full NAMAT model achieved the highest average AUPRC. Excluding graph-based message passing resulted in the largest performance decline, with average AUPRC decreasing from 0.683 to 0.516. Incorporating residual connections yielded a substantial improvement, increasing average AUPRC from 0.590 to 0.683. The input MLP consistently enhanced performance, and message-aware attention provided further average gains compared to uniform graph weighting in all three controlled comparisons. In the full model, replacing message-aware attention with uniform averaging reduced average AUPRC from 0.683 to 0.667, and NAMAT achieved higher AUPRC in 11 of the 12 cancer types, with UCEC as the exception.

**Table S8. Comprehensive ablation study across 12 cancer types.** Each block presents a controlled comparison isolating the effect of a specific component. `MAA`: message-aware attention; `MLP`: input feature transformation; `Res`: residual connection; `MSG`: graph-specific message information. Results are reported as mean AUPRC ± standard deviation over repeated runs, rounded to three decimals.

**Panel A. BRCA, BLCA, LUAD, LIHC, THCA, and LUSC.**

| Test | Configuration | BRCA | BLCA | LUAD | LIHC | THCA | LUSC |
|---|---|---|---|---|---|---|---|
| MAA | −MAA −MLP −Res | 0.670±0.006 | 0.616±0.008 | 0.499±0.012 | 0.462±0.023 | 0.315±0.014 | **0.385**±0.041 |
| | +MAA −MLP −Res | **0.736**±0.008 | **0.660**±0.013 | **0.559**±0.017 | **0.498**±0.027 | **0.327**±0.015 | 0.365±0.036 |
| MAA | −MAA +MLP −Res | 0.716±0.006 | 0.667±0.006 | 0.583±0.011 | 0.505±0.018 | 0.359±0.017 | **0.490**±0.023 |
| | +MAA +MLP −Res | **0.750**±0.005 | **0.692**±0.013 | **0.615**±0.011 | **0.515**±0.014 | **0.364**±0.018 | 0.462±0.031 |
| MAA | −MAA +MLP +Res | 0.771±0.004 | 0.774±0.007 | 0.724±0.006 | 0.593±0.010 | 0.412±0.022 | 0.590±0.047 |
| | +MAA +MLP +Res | **0.802**±0.006 | **0.789**±0.005 | **0.744**±0.008 | **0.595**±0.019 | **0.426**±0.026 | **0.616**±0.053 |
| MLP | −MAA −MLP −Res | 0.670±0.006 | 0.616±0.008 | 0.499±0.012 | 0.462±0.023 | 0.315±0.014 | 0.385±0.041 |
| | −MAA +MLP −Res | **0.716**±0.006 | **0.667**±0.006 | **0.583**±0.011 | **0.505**±0.018 | **0.359**±0.017 | **0.490**±0.023 |
| MLP | +MAA −MLP −Res | 0.736±0.008 | 0.660±0.013 | 0.559±0.017 | 0.498±0.027 | 0.327±0.015 | 0.365±0.036 |
| | +MAA +MLP −Res | **0.750**±0.005 | **0.692**±0.013 | **0.615**±0.011 | **0.515**±0.014 | **0.364**±0.018 | **0.462**±0.031 |
| Res | +MAA +MLP −Res | 0.750±0.005 | 0.692±0.013 | 0.615±0.011 | 0.515±0.014 | 0.364±0.018 | 0.462±0.031 |
| | +MAA +MLP +Res | **0.802**±0.006 | **0.789**±0.005 | **0.744**±0.008 | **0.595**±0.019 | **0.426**±0.026 | **0.616**±0.053 |
| MSG | −MSG +MLP +Res | 0.513±0.004 | 0.653±0.009 | 0.582±0.006 | 0.440±0.010 | 0.233±0.014 | 0.412±0.025 |
| | +MSG +MLP +Res | **0.802**±0.006 | **0.789**±0.005 | **0.744**±0.008 | **0.595**±0.019 | **0.426**±0.026 | **0.616**±0.053 |

**Panel B. ESCA, PRAD, STAD, COAD, UCEC, and CESC.**

| Test | Configuration | ESCA | PRAD | STAD | COAD | UCEC | CESC |
|---|---|---|---|---|---|---|---|
| MAA | −MAA −MLP −Res | 0.403±0.012 | 0.639±0.013 | 0.570±0.015 | 0.449±0.014 | **0.523**±0.014 | 0.498±0.039 |
| | +MAA −MLP −Res | **0.448**±0.023 | **0.665**±0.021 | **0.624**±0.009 | **0.456**±0.019 | 0.522±0.010 | **0.587**±0.062 |
| MAA | −MAA +MLP −Res | 0.479±0.014 | 0.726±0.019 | 0.609±0.013 | 0.497±0.007 | 0.560±0.014 | 0.643±0.047 |
| | +MAA +MLP −Res | **0.506**±0.019 | **0.747**±0.022 | **0.646**±0.010 | **0.508**±0.008 | **0.596**±0.015 | **0.677**±0.050 |
| MAA | −MAA +MLP +Res | 0.589±0.014 | 0.824±0.014 | 0.765±0.010 | 0.608±0.005 | **0.747**±0.007 | 0.610±0.037 |
| | +MAA +MLP +Res | **0.611**±0.015 | **0.827**±0.009 | **0.779**±0.009 | **0.610**±0.008 | 0.743±0.014 | **0.655**±0.030 |
| MLP | −MAA −MLP −Res | 0.403±0.012 | 0.639±0.013 | 0.570±0.015 | 0.449±0.014 | 0.523±0.014 | 0.498±0.039 |
| | −MAA +MLP −Res | **0.479**±0.014 | **0.726**±0.019 | **0.609**±0.013 | **0.497**±0.007 | **0.560**±0.014 | **0.643**±0.047 |
| MLP | +MAA −MLP −Res | 0.448±0.023 | 0.665±0.021 | 0.624±0.009 | 0.456±0.019 | 0.522±0.010 | 0.587±0.062 |
| | +MAA +MLP −Res | **0.506**±0.019 | **0.747**±0.022 | **0.646**±0.010 | **0.508**±0.008 | **0.596**±0.015 | **0.677**±0.050 |
| Res | +MAA +MLP −Res | 0.506±0.019 | 0.747±0.022 | 0.646±0.010 | 0.508±0.008 | 0.596±0.015 | **0.677**±0.050 |
| | +MAA +MLP +Res | **0.611**±0.015 | **0.827**±0.009 | **0.779**±0.009 | **0.610**±0.008 | **0.743**±0.014 | 0.655±0.030 |
| MSG | −MSG +MLP +Res | 0.456±0.011 | 0.579±0.014 | 0.676±0.006 | 0.526±0.006 | 0.678±0.009 | 0.447±0.032 |
| | +MSG +MLP +Res | **0.611**±0.015 | **0.827**±0.009 | **0.779**±0.009 | **0.610**±0.008 | **0.743**±0.014 | **0.655**±0.030 |

**Table S9.** AUPRC of NAMAT and alternative multi-graph fusion strategies across 12 cancer types (mean ± SD over 10 runs). Bold marks the highest mean. Superscripts compare NAMAT with Uniform, Global, PPI Concat, and Concat Attention, respectively: $^{*}$ indicates $p < 0.05$ and $^{-}$ indicates $p \geq 0.05$ (two-sided Wilcoxon signed-rank tests). Per-cancer tests used a separate repetition of 10 paired runs; displayed estimates are from the original experiments. Across-cancer comparisons give $p = 0.003$, 0.004, 0.002, and 0.001, respectively.

| Cancer | NAMAT | Uniform | Global | PPI Concat | Concat Attention |
|---|---|---|---|---|---|
| **BRCA** | **0.802±0.006**$^{(*,\,*,\,*,\,*)}$ | 0.771±0.004 | 0.775±0.003 | 0.786±0.010 | 0.782±0.006 |
| **BLCA** | **0.789±0.005**$^{(*,\,*,\,*,\,*)}$ | 0.774±0.007 | 0.775±0.005 | 0.768±0.007 | 0.772±0.007 |
| **LUAD** | **0.744±0.008**$^{(*,\,*,\,*,\,*)}$ | 0.724±0.006 | 0.729±0.008 | 0.716±0.012 | 0.720±0.005 |
| **LIHC** | **0.595±0.019**$^{(-,-,-,*)}$ | 0.593±0.010 | 0.590±0.020 | 0.590±0.017 | 0.571±0.013 |
| **THCA** | **0.426±0.026**$^{(*,\,*,\,*,\,*)}$ | 0.412±0.022 | 0.413±0.019 | 0.384±0.029 | 0.367±0.015 |
| **LUSC** | **0.616±0.053**$^{(-,-,*,\,*)}$ | 0.590±0.047 | 0.588±0.039 | 0.541±0.043 | 0.532±0.048 |
| **ESCA** | **0.611±0.015**$^{(*,\,*,\,*,\,*)}$ | 0.589±0.014 | 0.595±0.017 | 0.577±0.025 | 0.559±0.012 |
| **PRAD** | 0.827±0.009$^{(-,-,*,\,*)}$ | 0.824±0.014 | **0.829±0.008** | 0.789±0.015 | 0.800±0.018 |
| **STAD** | **0.779±0.009**$^{(*,\,*,\,*,-)}$ | 0.765±0.010 | 0.764±0.007 | 0.762±0.010 | 0.771±0.017 |
| **COAD** | **0.610±0.008**$^{(-,-,-,*)}$ | 0.608±0.005 | 0.608±0.005 | 0.606±0.009 | 0.598±0.007 |
| **UCEC** | 0.743±0.014$^{(-,-,-,-)}$ | 0.747±0.007 | 0.745±0.011 | **0.751±0.010** | 0.743±0.012 |
| **CESC** | **0.655±0.030**$^{(*,\,*,\,*,\,*)}$ | 0.610±0.037 | 0.617±0.027 | 0.602±0.041 | 0.598±0.027 |
| **Average** | **0.683** | 0.667 | 0.669 | 0.656 | 0.651 |

Table S10. Hits@K of NAMAT and alternative multi-graph fusion strategies across 12 cancer types for $K \in \{1,3,5,10,20,50\}$. All variants use the training protocol of NAMAT. Uniform averages graph-specific messages over the valid PPI networks; Global uses learned graph-level weights shared across all genes; PPI Concat processes each PPI network independently and concatenates the six final node embeddings before prediction; Concat Attention uses the same network-specific branches as PPI Concat but combines the six final node embeddings by a gene-specific weighted sum, with weights produced by an MLP from their concatenation. Values are means over 10 repeated runs, with each run averaged over five folds.

| $K$ | Method | BRCA | BLCA | LUAD | LIHC | THCA | LUSC | ESCA | PRAD | STAD | COAD | UCEC | CESC | Avg. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | **NAMAT** | 0.46 | 0.36 | 0.28 | 0.26 | 0.40 | 0.20 | 0.86 | 0.28 | 0.28 | 0.94 | 0.44 | 0.20 | **0.41** |
| | **Uniform** | 0.44 | 0.26 | 0.22 | 0.28 | 0.50 | 0.20 | 0.74 | 0.20 | 0.20 | 0.98 | 0.38 | 0.20 | 0.38 |
| | **Global** | 0.40 | 0.30 | 0.16 | 0.28 | 0.50 | 0.20 | 0.80 | 0.22 | 0.28 | 0.96 | 0.38 | 0.20 | 0.39 |
| | **PPI Concat** | 0.30 | 0.26 | 0.14 | 0.22 | 0.40 | 0.20 | 0.62 | 0.28 | 0.18 | 0.88 | 0.38 | 0.18 | 0.34 |
| | **Concat Attention** | 0.24 | 0.40 | 0.20 | 0.32 | 0.34 | 0.20 | 0.48 | 0.28 | 0.42 | 0.90 | 0.62 | 0.18 | 0.38 |
| 3 | **NAMAT** | 1.08 | 1.74 | 0.72 | 0.80 | 0.68 | 0.36 | 1.60 | 0.82 | 1.26 | 2.14 | 1.22 | 0.26 | **1.06** |
| | **Uniform** | 1.14 | 1.40 | 0.60 | 0.66 | 0.66 | 0.38 | 1.56 | 0.70 | 1.00 | 2.22 | 1.12 | 0.22 | 0.97 |
| | **Global** | 1.04 | 1.44 | 0.62 | 0.66 | 0.68 | 0.38 | 1.66 | 0.68 | 0.96 | 2.22 | 1.10 | 0.24 | 0.97 |
| | **PPI Concat** | 0.94 | 1.30 | 0.42 | 0.74 | 0.62 | 0.30 | 1.38 | 0.60 | 1.00 | 2.12 | 1.22 | 0.26 | 0.91 |
| | **Concat Attention** | 0.98 | 1.64 | 0.70 | 0.82 | 0.44 | 0.40 | 1.08 | 0.74 | 1.18 | 2.04 | 1.14 | 0.20 | 0.95 |
| 5 | **NAMAT** | 1.86 | 2.70 | 1.00 | 0.96 | 0.80 | 0.44 | 2.08 | 1.06 | 1.46 | 2.58 | 1.68 | 0.68 | **1.44** |
| | **Uniform** | 1.94 | 1.98 | 0.94 | 0.82 | 0.82 | 0.44 | 2.06 | 1.10 | 1.28 | 2.74 | 1.78 | 0.58 | 1.37 |
| | **Global** | 1.96 | 2.02 | 0.94 | 0.84 | 0.90 | 0.44 | 2.18 | 1.02 | 1.36 | 2.74 | 1.80 | 0.64 | 1.40 |
| | **PPI Concat** | 1.58 | 2.00 | 0.60 | 0.90 | 0.84 | 0.38 | 1.72 | 0.80 | 1.38 | 2.60 | 1.76 | 0.62 | 1.27 |
| | **Concat Attention** | 1.62 | 2.64 | 1.00 | 1.00 | 0.56 | 0.44 | 1.36 | 0.98 | 1.40 | 2.40 | 1.68 | 0.32 | 1.28 |
| 10 | **NAMAT** | 3.66 | 4.44 | 1.48 | 1.18 | 1.04 | 0.56 | 2.72 | 2.12 | 2.14 | 3.22 | 2.76 | 1.10 | **2.20** |
| | **Uniform** | 3.40 | 3.42 | 1.42 | 1.24 | 1.04 | 0.54 | 2.58 | 1.74 | 1.90 | 3.16 | 2.62 | 1.10 | 2.01 |
| | **Global** | 3.36 | 3.60 | 1.44 | 1.20 | 1.04 | 0.56 | 2.74 | 1.72 | 1.94 | 3.20 | 2.78 | 1.14 | 2.06 |
| | **PPI Concat** | 3.08 | 3.62 | 1.18 | 1.26 | 0.96 | 0.50 | 2.30 | 1.66 | 2.14 | 3.16 | 2.34 | 1.10 | 1.94 |
| | **Concat Attention** | 3.34 | 4.12 | 1.50 | 1.26 | 0.70 | 0.56 | 1.72 | 1.96 | 2.16 | 3.12 | 2.68 | 0.88 | 2.00 |
| 20 | **NAMAT** | 4.52 | 5.90 | 2.28 | 1.72 | 1.26 | 0.70 | 3.36 | 2.70 | 3.24 | 4.12 | 3.38 | 1.20 | **2.87** |
| | **Uniform** | 5.26 | 4.90 | 2.30 | 1.76 | 1.24 | 0.68 | 3.16 | 2.80 | 2.86 | 3.88 | 3.40 | 1.20 | 2.79 |
| | **Global** | 5.12 | 5.06 | 2.32 | 1.64 | 1.30 | 0.68 | 3.20 | 2.82 | 2.86 | 3.96 | 3.42 | 1.20 | 2.80 |
| | **PPI Concat** | 4.62 | 4.80 | 1.78 | 1.70 | 1.18 | 0.62 | 2.88 | 2.78 | 3.20 | 3.76 | 3.10 | 1.20 | 2.64 |
| | **Concat Attention** | 4.64 | 5.56 | 2.54 | 1.72 | 1.08 | 0.62 | 2.54 | 2.78 | 3.16 | 4.00 | 3.44 | 1.12 | 2.77 |
| 50 | **NAMAT** | 8.10 | 7.30 | 4.78 | 3.52 | 1.60 | 1.00 | 3.94 | 4.26 | 4.82 | 5.38 | 4.76 | 1.34 | **4.23** |
| | **Uniform** | 7.72 | 6.94 | 4.68 | 3.42 | 1.58 | 0.92 | 3.90 | 4.42 | 4.86 | 5.44 | 4.80 | 1.26 | 4.16 |
| | **Global** | 7.82 | 7.04 | 4.24 | 3.36 | 1.60 | 0.98 | 3.90 | 4.40 | 4.92 | 5.42 | 4.82 | 1.22 | 4.14 |
| | **PPI Concat** | 7.48 | 6.96 | 4.32 | 3.20 | 1.50 | 0.86 | 3.60 | 4.18 | 4.36 | 5.22 | 4.82 | 1.22 | 3.98 |
| | **Concat Attention** | 7.90 | 7.10 | 4.90 | 3.22 | 1.52 | 0.92 | 3.74 | 4.24 | 4.60 | 5.54 | 4.38 | 1.20 | 4.11 |

## 2.5 Biological relevance analysis

To determine whether the top-ranked NAMAT predictions represent biologically meaningful signals, three complementary evaluation strategies were employed: over-representation analysis (ORA), pathway-level consistency analysis using gene set enrichment analysis (GSEA), and validation against external disease-evidence databases.

**Functional enrichment analysis.** ORA was first conducted on the top-ranked genes for CESC, LUSC, and PRAD using Human Disease Ontology (HDO) and KEGG gene sets. This analysis evaluates whether disease-relevant gene sets are statistically over-represented among the top-ranked predictions.

**Table S11.** ORA of top-K predicted genes (K= 100–200) using HDO and KEGG. Count denotes the number of overlapping genes, FE denotes fold enrichment, and FDR was computed using the Benjamini-Hochberg correction.

| Cancer | Term | Collection | K | Count | FE | FDR (BH) |
|---|---|---|---|---|---|---|
| CESC | Cervical cancer | HDO | 100 | 9 | 7.03 | $3.87 \times 10^{-5}$ |
| | | | 150 | 13 | 6.63 | $7.39 \times 10^{-7}$ |
| | | | 200 | 16 | 6.25 | $4.86 \times 10^{-8}$ |
| | hsa05165 | KEGG | 100 | 15 | 3.48 | $1.18 \times 10^{-4}$ |
| | | | 150 | 19 | 2.99 | $8.77 \times 10^{-5}$ |
| | | | 200 | 27 | 3.23 | $3.64 \times 10^{-7}$ |
| LUSC | Lung SCC | HDO | 100 | 3 | 4.53 | $9.01 \times 10^{-2}$ |
| | | | 150 | 3 | 2.99 | $1.66 \times 10^{-1}$ |
| | | | 200 | 5 | 3.75 | $3.97 \times 10^{-2}$ |
| | hsa05223 | KEGG | 100 | 10 | 10.04 | $7.17 \times 10^{-7}$ |
| | | | 150 | 12 | 8.17 | $2.67 \times 10^{-7}$ |
| | | | 200 | 15 | 7.77 | $6.63 \times 10^{-9}$ |
| PRAD | Prostate cancer | HDO | 100 | 20 | 4.79 | $3.67 \times 10^{-7}$ |
| | | | 150 | 29 | 4.63 | $2.52 \times 10^{-10}$ |
| | | | 200 | 37 | 4.43 | $4.56 \times 10^{-13}$ |
| | hsa05215 | KEGG | 100 | 10 | 8.91 | $2.35 \times 10^{-6}$ |
| | | | 150 | 14 | 8.16 | $1.86 \times 10^{-8}$ |
| | | | 200 | 19 | 8.41 | $1.05 \times 10^{-11}$ |

Supplementary Table S11 demonstrates that NAMAT predictions are enriched for disease-relevant Human Disease Ontology (HDO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) terms. Fold enrichment values ranged from 2.99 to 10.04, with the majority of results exhibiting strong statistical significance. Prostate adenocarcinoma (PRAD) displayed consistent enrichment for both prostate cancer HDO and KEGG terms, whereas lung squamous cell carcinoma (LUSC) exhibited particularly strong KEGG enrichment. In cervical squamous cell carcinoma and endocervical adenocarcinoma (CESC), enrichment of the HPV-related KEGG pathway hsa05165 aligns with the established biological role of HPV-associated processes in cervical cancer.

**Pathway-level consistency analysis.** The full ranked gene list was subsequently evaluated using Gene Set Enrichment Analysis (GSEA). In contrast to Over-Representation Analysis (ORA), which considers only a fixed top-K subset, GSEA assesses whether genes from predefined disease or pathway gene sets are concentrated near the top of the complete ranked list.

Supplementary Table S12 demonstrates that disease-associated gene sets are strongly enriched near the top of the NAMAT-ranked gene lists. All evaluated terms yielded positive normalized enrichment score (NES) values, with NES reaching up to 4.29 and highly significant false discovery rate (FDR) values. The leading-edge fractions indicate that disease-relevant genes contribute substantially to the enrichment signals, thereby supporting the ranking-level biological consistency of NAMAT predictions.

**Comparison with external databases.** Top-ranked predictions were cross-checked against external disease-gene association databases. This analysis assessed whether genes prioritized by NAMAT overlap with known disease-associated genes from resources not used as model labels.

**Table S12.** Gene Set Enrichment Analysis (GSEA) of the ranked gene list using Human Disease Ontology (HDO) and Kyoto Encyclopedia of Genes and Genomes (KEGG). NES denotes normalized enrichment score, and FDR was computed using the Benjamini–Hochberg correction. Leading-edge (LE) values are reported as tags/list percentages. Tags is the percentage of genes in the set that fall within the leading edge, and list is the percentage of the ranked list reached at the enrichment peak. For example, 38/6 means that 38% of the gene set appears within the top 6% of the ranked list.

| Cancer | Term/Pathway | Collection | Set size | NES | FDR (BH) | LE (tags/list, %) |
|---|---|---|---|---|---|---|
| **CESC** | Cervical SCC | HDO | 21 | 2.65 | $1.66 \times 10^{-4}$ | 38/6 |
| | hsa05165 | KEGG | 290 | 4.29 | $3.54 \times 10^{-48}$ | 48/13 |
| **LUSC** | Lung SCC | HDO | 46 | 2.22 | $2.51 \times 10^{-5}$ | 35/12 |
| | hsa05223 | KEGG | 65 | 3.49 | $1.91 \times 10^{-21}$ | 80/20 |
| **PRAD** | Prostate cancer | HDO | 286 | 3.54 | $4.71 \times 10^{-66}$ | 61/16 |
| | hsa05215 | KEGG | 84 | 3.68 | $4.02 \times 10^{-33}$ | 74/14 |

**Table S13.** DisGeNET validation of top-K predicted genes (K=50). Count denotes the number of overlapping genes, P@K denotes precision at K, FE denotes fold enrichment, and FDR was computed using Benjamini–Hochberg correction.

| Cancer | K | Count | P@K | FE | FDR (BH) |
|---|---|---|---|---|---|
| **CESC** | 50 | 2 | 0.04 | 16.32 | $1.32 \times 10^{-2}$ |
| **LUSC** | 50 | 3 | 0.06 | 13.43 | $1.89 \times 10^{-3}$ |
| **PRAD** | 50 | 9 | 0.18 | 4.91 | $9.56 \times 10^{-5}$ |

**Table S14.** Open Targets validation of top-K predicted genes (K=50, score $\geq 0.20$). Count denotes the number of overlapping genes, P@K denotes precision at K, FE denotes fold enrichment, and FDR was computed using Benjamini–Hochberg correction.

| Cancer | Threshold | K | Count | P@K | FE | FDR (BH) |
|---|---|---|---|---|---|---|
| **CESC** | 0.20 | 50 | 25 | 0.50 | 6.78 | $1.02 \times 10^{-15}$ |
| **LUSC** | 0.20 | 50 | 16 | 0.32 | 6.45 | $1.48 \times 10^{-9}$ |
| **PRAD** | 0.20 | 50 | 13 | 0.26 | 5.70 | $3.17 \times 10^{-7}$ |

Supplementary Tables S13 and S14 demonstrate that the top-ranked NAMAT predictions overlap significantly with established disease-associated genes. In DisGeNET, CESC and LUSC exhibit high fold enrichment, although the absolute overlap counts remain low, likely due to limited disease annotations. In contrast, Open Targets displays higher overlap counts, with 25 CESC-associated genes, 16 LUSC-associated genes, and 13 PRAD-associated genes identified among the top 50 predictions at a score threshold of $\geq 0.20$.

Together, the enrichment and disease-association analyses show that NAMAT predictions align with external cancer-related evidence. These findings support their biological coherence but do not establish independence from benchmark-label information or confirm individual predictions as cancer drivers.

### 2.6 Interpretation of node-wise graph attention in NAMAT

To elucidate the prediction mechanism of NAMAT, this study investigates how the model integrates information from multiple PPI networks during message passing. Three complementary aspects of the learned model behavior are analyzed: (i) gene-specific graph-attention patterns; (ii) attention-weighted message magnitudes; and (iii) neighbor-level scores. Attention coefficients indicate the networks emphasized by the trained model rather than independently measured biological reliability. We analyze the four representative STAD genes from the main manuscript (PIK3CA, APC, MYH9, and TP53) and four automatically selected high-ranked unlabeled candidates (PCDH10, ACTN2, FBN1, and PRKDC).

The primary analysis uses the first message-passing layer, matching Figures 3–5 of the main manuscript. All explanations are obtained in evaluation mode, with PPI dropout disabled and attention temperature $\tau = 0.6$. For each known gene, we use the model from the fold in which that gene was held out in each of the 10 repeated runs. Unlabeled candidates are never labeled training examples; therefore, all 50 models from five-fold cross-validation repeated 10 times are used. Candidate explanations are averaged over five folds within each run when quantifying stability.

#### 2.6.1 Gene-specific graph attention patterns




**Figure S1.** Graph attention coefficients learned by NAMAT for representative driver genes in stomach adenocarcinoma (STAD). Bars represent mean attention coefficients averaged across 10 repeated runs. Results use the first message-passing layer and the held-out model for each gene in each run.

The analysis first assessed whether NAMAT learns gene-specific preferences across interaction networks by examining the graph attention coefficients assigned to representative driver genes in stomach adenocarcinoma (STAD). Figure S1 presents the mean attention coefficients learned for PIK3CA, APC, MYH9, and TP53. The attention distributions differ substantially across genes, suggesting that NAMAT does not apply a fixed global weighting to PPI networks. PIK3CA most strongly emphasizes CPDB, APC emphasizes STRING, MYH9 emphasizes MultiNet, and TP53 emphasizes PCNet. Detailed first-layer profiles for both the known genes and unlabeled candidates are provided in Table S16.

#### 2.6.2 Network contribution analysis

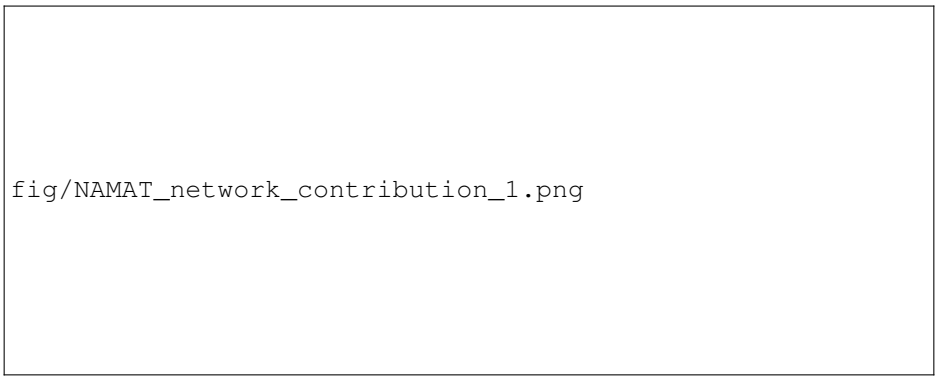



**Figure S2.** Network contribution scores learned by NAMAT for representative driver genes in stomach adenocarcinoma (STAD). Bars represent mean contribution magnitudes averaged across 10 repeated runs. Results use the first message-passing layer and the held-out model for each gene in each run.

The subsequent analysis examined how information from different interaction networks contributes to node representation learning. For each gene and PPI network, the network contribution was summarized as the magnitude of the graph-specific message after attention weighting. Specifically, $C_{i,k} = \|\alpha_{i,k} m_{i,k}\|_2$, where $\alpha_{i,k}$ is the attention coefficient and $m_{i,k}$ is the graph-specific message. Contribution magnitudes therefore depend on both attention and message strength; the network receiving the highest attention need not provide the largest weighted message.

Figure S2 presents the mean contribution magnitudes associated with PIK3CA, APC, MYH9, and TP53. These profiles provide complementary descriptions of the information entering aggregation. They characterize the trained model's representations rather than causal effects of individual networks on its predictions.

#### 2.6.3 Neighbor-level evidence supporting driver gene prediction




**Figure S3.** Top scoring neighbors identified by NAMAT for representative driver genes in stomach adenocarcinoma (STAD). Node size is proportional to the mean neighbor score, averaged across 10 repeated runs. The top two neighbors are selected separately within each PPI using first-layer explanations from held-out models.

To further investigate the evidence utilized by NAMAT during prediction, neighbors with the highest model-derived scores were analyzed. Figure S3 presents the selected neighbors identified from the six PPI networks, with node size proportional to the corresponding neighbor score.

For target gene $i$, PPI $k$, and neighbor $j$, the score is $s_{i,k,j} = \alpha_{i,k}\|h_j\|_2$, with self-neighbors excluded. Scores are averaged across the relevant models before ranking neighbors separately within each PPI. The attention coefficient is common to all neighbors of a given target within a PPI in an individual model, so their relative ranking in that model is determined by representation norms. The score does not measure the effect of removing a neighbor on the prediction.

Several meaningful patterns emerge from the analysis. For instance, ERBB3, KRAS, and TP53 are among the highest-scoring neighbors for PIK3CA, while TP53, PIK3CA, and CDH1 are highlighted for APC. Additionally, recurrent neighbors such as TTN, TP53, ARID1A, and PTEN are identified for MYH9 and TP53 across multiple interaction networks.

These neighbors represent model-derived associations rather than direct causal mechanisms. We assess their reproducibility and external biological support below.

#### 2.6.4 Selection and interpretation of unlabeled STAD candidates

The 11,367 unlabeled STAD genes, defined as genes that are neither benchmark positives nor benchmark negatives, were ranked by their mean predicted score across all 50 cross-validation models. A predefined novelty filter excluded genes listed in NCG 6.1, COSMIC Cancer Gene Census v104, 28 KEGG cancer pathways (hsa052xx), cancer-related OMIM entries (genemap2 2019, mapping key 3), Open Targets stomach-cancer associations (MONDO_0005036, release 26.09, score $\geq 0.20$), or curated DisGeNET stomach-cancer associations (C0024623, score $\geq 0.40$). The combined resource list contained 3,640 genes. The four highest-ranked remaining unlabeled genes were selected automatically, without inspecting their interpretation results.

Table S15 reports the selected candidates and their ranking stability. All four were absent from the specified resource lists at the stated thresholds. This operational selection rule does not establish that the genes have no previous cancer-related literature. Prediction scores are uncalibrated model outputs used for prioritization and do not constitute validated probabilities of driver status.

Candidate attention and contribution profiles are reported alongside the known genes in Table S16. PCDH10 emphasized IRefIndex and PCNet in attention, while PCNet provided its largest mean contribution magnitude. ACTN2 emphasized MultiNet in attention but PCNet in contribution. STRING had the largest mean attention and contribution for FBN1, whereas PCNet had the largest values for PRKDC. PCDH10 had no neighbors in STRING or IRefIndex_2015 and only one neighbor in

Table S15. High-ranked unlabeled STAD candidates selected automatically after the predefined resource filter. Overall rank is among all 11,367 unlabeled genes before filtering. Scores are means across 50 models; SD is calculated across 10 run-level means, each averaged over five folds. Median, best, and worst ranks are calculated across the same 10 runs.

| Gene | Overall rank | Mean score ± SD | Median rank | Best rank | Worst rank |
|---|---|---|---|---|---|
| **PCDH10** | 13 | 0.9989 ± 0.0011 | 11.5 | 10 | 13 |
| **ACTN2** | 29 | 0.9932 ± 0.0030 | 28.0 | 27 | 32 |
| **FBN1** | 33 | 0.9919 ± 0.0035 | 33.5 | 25 | 45 |
| **PRKDC** | 34 | 0.9917 ± 0.0043 | 35.5 | 24 | 43 |

MultiNet. Networks without neighbors receive zero attention and are displayed as unavailable in the profile table.

Table S16. First-layer graph-attention coefficients and network contribution magnitudes for four known STAD genes and four unlabeled candidates. Values are mean ± SD across 10 runs. Known genes use one held-out model per run; candidates use the mean of five fold models within each run. Contribution is $C_{i,k} = \|\alpha_{i,k} m_{i,k}\|_2$. A dash indicates that the gene has no neighbors in the PPI.

**Panel A. Graph-attention coefficients.**

| Gene | CPDB | STRING | MultiNet | PCNet | IRefIndex | IRefIndex2015 |
|---|---|---|---|---|---|---|
| **PIK3CA** | 0.472 ± 0.182 | 0.043 ± 0.039 | 0.124 ± 0.040 | 0.181 ± 0.108 | 0.107 ± 0.042 | 0.073 ± 0.037 |
| **APC** | 0.039 ± 0.020 | 0.605 ± 0.157 | 0.002 ± 0.002 | 0.331 ± 0.125 | 0.022 ± 0.014 | 0.002 ± 0.002 |
| **MYH9** | 0.041 ± 0.045 | 0.087 ± 0.036 | 0.603 ± 0.215 | 0.109 ± 0.088 | 0.051 ± 0.044 | 0.109 ± 0.034 |
| **TP53** | 0.035 ± 0.029 | 0.056 ± 0.036 | 0.049 ± 0.031 | 0.775 ± 0.157 | 0.052 ± 0.035 | 0.034 ± 0.027 |
| **PCDH10** | 0.213 ± 0.010 | – | 0.007 ± 0.007 | 0.388 ± 0.033 | 0.391 ± 0.026 | – |
| **ACTN2** | 0.081 ± 0.010 | 0.143 ± 0.011 | 0.342 ± 0.058 | 0.265 ± 0.030 | 0.089 ± 0.010 | 0.081 ± 0.008 |
| **FBN1** | 0.075 ± 0.009 | 0.444 ± 0.038 | 0.051 ± 0.008 | 0.148 ± 0.023 | 0.187 ± 0.004 | 0.094 ± 0.008 |
| **PRKDC** | 0.134 ± 0.007 | 0.249 ± 0.032 | 0.081 ± 0.008 | 0.313 ± 0.031 | 0.153 ± 0.004 | 0.070 ± 0.009 |

**Panel B. Network contribution magnitudes.**

| Gene | CPDB | STRING | MultiNet | PCNet | IRefIndex | IRefIndex2015 |
|---|---|---|---|---|---|---|
| **PIK3CA** | 1.432 ± 0.655 | 0.240 ± 0.191 | 0.454 ± 0.111 | 0.936 ± 0.612 | 0.514 ± 0.189 | 0.297 ± 0.094 |
| **APC** | 0.201 ± 0.100 | 1.780 ± 0.677 | 0.014 ± 0.016 | 1.359 ± 0.578 | 0.131 ± 0.083 | 0.014 ± 0.016 |
| **MYH9** | 0.267 ± 0.284 | 0.345 ± 0.128 | 0.977 ± 0.412 | 0.493 ± 0.401 | 0.286 ± 0.239 | 0.383 ± 0.111 |
| **TP53** | 0.301 ± 0.219 | 0.452 ± 0.254 | 0.390 ± 0.223 | 5.207 ± 1.585 | 0.430 ± 0.251 | 0.282 ± 0.198 |
| **PCDH10** | 0.634 ± 0.045 | – | 0.042 ± 0.036 | 1.291 ± 0.119 | 1.021 ± 0.142 | – |
| **ACTN2** | 0.469 ± 0.047 | 0.695 ± 0.056 | 1.106 ± 0.262 | 1.297 ± 0.160 | 0.500 ± 0.044 | 0.442 ± 0.030 |
| **FBN1** | 0.292 ± 0.019 | 1.257 ± 0.175 | 0.185 ± 0.025 | 0.677 ± 0.097 | 0.629 ± 0.040 | 0.341 ± 0.018 |
| **PRKDC** | 0.497 ± 0.025 | 0.668 ± 0.126 | 0.300 ± 0.013 | 1.151 ± 0.143 | 0.533 ± 0.031 | 0.279 ± 0.016 |

Table S17 lists the highest-scoring candidate neighbors within each PPI. Examples include FAT4 and CDH1 for PCDH10 in PCNet, TTN and PIK3CA for ACTN2 in STRING and PCNet, and TP53 across all six networks for PRKDC. These examples describe model-derived neighborhood patterns; statistical support is assessed using all neighbors rather than only the displayed top two.

Figures S4–S6 provide corresponding visualizations of the candidate attention profiles, network contributions, and top neighbors.

Table S17. Top neighbors of the four unlabeled STAD candidates within each PPI, using first-layer scores averaged across 50 models. Each entry gives the neighbor, mean score, and number of runs in which it appeared in the top two, in that order. Run-level rankings use scores averaged across five folds. A dash indicates no available neighbor in that position. PCDH10 has no neighbors in STRING or IRefIndex_2015 and one neighbor in MultiNet.

| Candidate | PPI | Top Neighbor 1: Score; Runs | Top Neighbor 2: Score; Runs |
|---|---|---|---|
| PCDH10 | CPDB | PCDH12: 0.3327; 10/10 | TMEM30B: 0.2569; 10/10 |
| PCDH10 | STRING | – | – |
| PCDH10 | MultiNet | CTCF: 0.0117; 10/10 | – |
| PCDH10 | PCNet | FAT4: 6.4455; 10/10 | CDH1: 5.4741; 10/10 |
| PCDH10 | IRefIndex | PCDH12: 0.6187; 10/10 | TMEM30B: 0.4770; 10/10 |
| PCDH10 | IRefIndex_2015 | – | – |
| ACTN2 | CPDB | TTN: 2.6571; 10/10 | DYSF: 0.3762; 10/10 |
| ACTN2 | STRING | TTN: 4.7685; 10/10 | PIK3CA: 2.1607; 10/10 |
| ACTN2 | MultiNet | GRIN2B: 1.0442; 10/10 | NRIP1: 0.7188; 10/10 |
| ACTN2 | PCNet | TTN: 8.8947; 10/10 | PIK3CA: 4.0304; 10/10 |
| ACTN2 | IRefIndex | TTN: 2.9289; 10/10 | DMD: 0.5632; 10/10 |
| ACTN2 | IRefIndex_2015 | LRP12: 0.2841; 10/10 | KCNN2: 0.2669; 10/10 |
| FBN1 | CPDB | VCAN: 0.2615; 10/10 | FBN2: 0.2316; 10/10 |
| FBN1 | STRING | APOB: 1.9019; 10/10 | CDH2: 1.7622; 10/10 |
| FBN1 | MultiNet | VCAN: 0.1732; 10/10 | FBLN2: 0.1028; 10/10 |
| FBN1 | PCNet | FAT4: 2.4092; 10/10 | SYNE1: 2.1370; 10/10 |
| FBN1 | IRefIndex | VCAN: 0.6722; 10/10 | FBN2: 0.5989; 10/10 |
| FBN1 | IRefIndex_2015 | VCAN: 0.3248; 10/10 | FBN2: 0.2884; 10/10 |
| PRKDC | CPDB | TP53: 3.6101; 10/10 | LRRK2: 0.9501; 10/10 |
| PRKDC | STRING | TP53: 6.9136; 10/10 | MDC1: 0.8783; 10/10 |
| PRKDC | MultiNet | TP53: 2.1698; 10/10 | ATM: 0.2078; 7/10 |
| PRKDC | PCNet | TP53: 8.5774; 10/10 | ARID1A: 7.2931; 10/10 |
| PRKDC | IRefIndex | TP53: 4.1386; 10/10 | LRRK2: 1.0893; 10/10 |
| PRKDC | IRefIndex_2015 | TP53: 1.8575; 10/10 | ATM: 0.1789; 7/10 |




Figure S4. First-layer graph-attention coefficients for four high-ranked unlabeled STAD candidates. Bars represent means across 10 runs, each averaged over five fold models. PCDH10 has no neighbors in STRING or IRefIndex_2015; these networks receive zero attention.

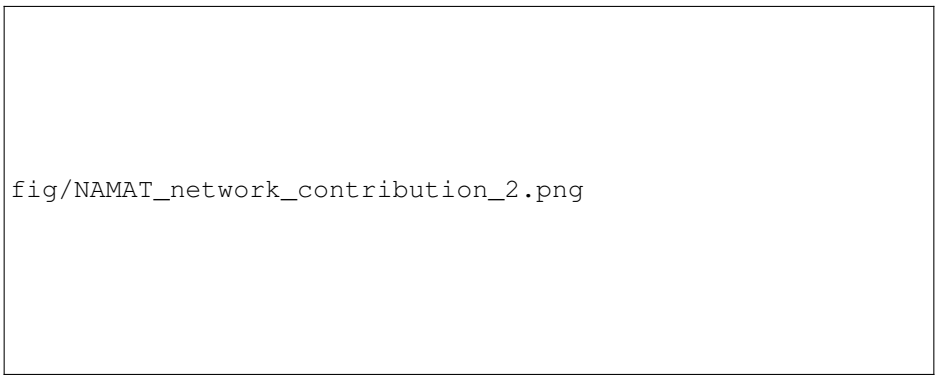



Figure S5. First-layer network contribution magnitudes for four high-ranked unlabeled STAD candidates. Contribution is defined as $C_{i,k} = \|\alpha_{i,k} m_{i,k}\|_2$. Bars represent means across 10 runs, each averaged over five fold models.




Figure S6. Top-scoring neighbors of four high-ranked unlabeled STAD candidates. The top two neighbors are selected separately within each PPI using first-layer scores averaged across 50 cross-validation models. Node area increases with the mean neighbor score on a common scale across all four panels, with a minimum area for readability. All available neighbors are shown when fewer than two exist; networks without neighbors are indicated explicitly.

#### 2.6.5 Quantitative stability of interpretation

Stability was calculated separately for each gene across 10 run-level explanations. For known genes, each explanation comes from one held-out model. For candidates, each explanation averages five fold models within the run. We calculated pairwise Spearman correlations of the PPI attention vectors and contribution vectors across all $\binom{10}{2} = 45$ pairs of runs. For PCDH10, which has no neighbors in STRING or IRefIndex_2015, Spearman correlations were computed over its four available networks. Top-PPI agreement records the most frequently selected highest-attention or highest-contribution PPI and the number of runs selecting it.

For neighbor stability, the top two neighbors within each PPI were encoded as $(\text{PPI}, \text{neighbor})$ pairs. If a PPI had fewer than two neighbors, all available neighbors were retained. Thus, each explanation contained 12 pairs for seven genes and seven pairs for PCDH10. For run-level sets $S_r$ and $S_s$, Jaccard similarity was calculated as

$$J(S_r, S_s) = \frac{|S_r \cap S_s|}{|S_r \cup S_s|}. \tag{9}$$

Means and SDs summarize the 45 pairwise comparisons descriptively; these comparisons are not independent because each run appears in multiple pairs.

Table S18. First-layer interpretation stability for known STAD genes and unlabeled candidates. Spearman and Jaccard values are mean ± SD across 45 run pairs. Top PPI gives the most frequent highest-weight network and its agreement across 10 runs. Known genes use one held-out model per run; candidates use five-fold averages within each run. Neighbor Jaccard uses 12 PPI–neighbor pairs per explanation, except PCDH10, which has seven.

| | Attention | | Contribution | | Neighbors |
|---|---|---|---|---|---|
| **Gene** | **Spearman** | **Top PPI** | **Spearman** | **Top PPI** | **Jaccard** |
| **Known genes** | | | | | |
| **PIK3CA** | 0.703 ± 0.259 | CPDB (9/10) | 0.671 ± 0.316 | CPDB (6/10) | 0.969 ± 0.062 |
| **APC** | 0.977 ± 0.031 | STRING (9/10) | 0.943 ± 0.042 | STRING (6/10) | 0.969 ± 0.062 |
| **MYH9** | 0.741 ± 0.418 | MultiNet (9/10) | 0.255 ± 0.742 | MultiNet (8/10) | 0.874 ± 0.107 |
| **TP53** | 0.915 ± 0.069 | PCNet (10/10) | 0.915 ± 0.086 | PCNet (10/10) | 1.000 ± 0.000 |
| Unlabeled candidates | | | | | |
| **PCDH10** | 0.893 ± 0.101 | IRefIndex (6/10) | 0.960 ± 0.081 | PCNet (9/10) | 1.000 ± 0.000 |
| **ACTN2** | 0.949 ± 0.046 | MultiNet (8/10) | 0.947 ± 0.046 | PCNet (7/10) | 1.000 ± 0.000 |
| **FBN1** | 1.000 ± 0.000 | STRING (10/10) | 0.968 ± 0.029 | STRING (10/10) | 1.000 ± 0.000 |
| **PRKDC** | 0.989 ± 0.023 | PCNet (9/10) | 0.939 ± 0.083 | PCNet (10/10) | 0.867 ± 0.144 |

Table S18 shows reproducible attention patterns and neighbor sets for the known genes. Contribution stability varied between genes: MYH9 had a low mean contribution correlation with substantial variation, although MultiNet remained its highest-contribution PPI in eight runs. Candidate explanations were also reproducible, but their five-fold averaging can smooth variation; their stability values should therefore not be interpreted as evidence of intrinsically greater stability than the known-gene explanations.

#### 2.6.6 Permutation tests of neighbor-level biological relevance

We evaluated whether the neighbor rankings preferentially prioritized externally supported STAD-associated genes. The reference set contained 815 genes from the union of Open Targets stomach-cancer associations (MONDO_0005036, release 26.09, score $\geq 0.20$) and curated DisGeNET stomach-cancer associations (C0024623, score $\geq 0.40$). Reference membership was used only to evaluate the neighbor rankings in this analysis.

For each target gene and PPI separately, all non-self neighbors were assigned their mean NAMAT neighbor score, averaged across the relevant models: 10 held-out models for known genes and 50 models for candidates. AUPRC was computed using these continuous scores and binary reference-set membership. The top-two neighbors were used for display only; statistical testing used the entire neighborhood.

Reference labels were randomly permuted among the same neighbors 10,000 times, preserving the number of reference

genes. The one-sided empirical $p$-value was calculated as

$$p_{g,k} = \frac{1 + \sum_{b=1}^{B} \mathrm{I}\left(\mathrm{AUPRC}_{g,k}^{(b)} \geq \mathrm{AUPRC}_{g,k}^{\mathrm{obs}}\right)}{B+1}, \qquad B = 10{,}000, \tag{10}$$

where $\mathrm{I}(\cdot)$ is the indicator function and $g,k$ identify the target gene and PPI. This tests whether reference genes are ranked unusually highly relative to randomly assigned reference labels within the same neighborhood. Networks are never pooled during permutation. Combinations with no neighbors, no reference neighbors, or exclusively reference neighbors were marked NA. Benjamini–Hochberg correction was applied jointly across all 44 evaluable first-layer tests; significance was defined as $q < 0.05$.

Table S19. First-layer neighbor-ranking permutation results for eight STAD genes across six PPIs. $N$ is the number of non-self neighbors and $R$ the number belonging to the external STAD reference set. Observed and null values are AUPRC; null values are means across 10,000 label permutations. BH correction was applied jointly across the 44 evaluable tests. An asterisk marks $q < 0.05$. NA indicates a non-testable combination.

| Gene | PPI | N | R | Observed | Null Mean | $p$ | $q$ |
|---|---|---|---|---|---|---|---|
| PIK3CA | CPDB | 96 | 38 | 0.5928 | 0.4225 | 0.0031 | 0.0062* |
| PIK3CA | STRING | 433 | 95 | 0.3639 | 0.2296 | 0.0001 | 0.0004* |
| PIK3CA | MultiNet | 41 | 15 | 0.6039 | 0.4187 | 0.0267 | 0.0392* |
| PIK3CA | PCNet | 1865 | 267 | 0.2489 | 0.1464 | 0.0001 | 0.0004* |
| PIK3CA | IRefIndex | 348 | 62 | 0.3362 | 0.1909 | 0.0002 | 0.0006* |
| PIK3CA | IRefIndex_2015 | 42 | 14 | 0.6045 | 0.3870 | 0.0123 | 0.0208* |
| APC | CPDB | 194 | 31 | 0.3040 | 0.1806 | 0.0036 | 0.0069* |
| APC | STRING | 115 | 17 | 0.6129 | 0.1796 | 0.0001 | 0.0004* |
| APC | MultiNet | 97 | 19 | 0.3490 | 0.2295 | 0.0263 | 0.0392* |
| APC | PCNet | 865 | 152 | 0.3359 | 0.1817 | 0.0001 | 0.0004* |
| APC | IRefIndex | 237 | 29 | 0.2910 | 0.1410 | 0.0008 | 0.0020* |
| APC | IRefIndex_2015 | 108 | 20 | 0.2983 | 0.2183 | 0.0697 | 0.0929 |
| MYH9 | CPDB | 328 | 44 | 0.2997 | 0.1487 | 0.0002 | 0.0006* |
| MYH9 | STRING | 68 | 6 | 0.2739 | 0.1409 | 0.0686 | 0.0929 |
| MYH9 | MultiNet | 22 | 4 | 0.7986 | 0.2880 | 0.0038 | 0.0070* |
| MYH9 | PCNet | 1314 | 155 | 0.2050 | 0.1224 | 0.0001 | 0.0004* |
| MYH9 | IRefIndex | 267 | 28 | 0.2725 | 0.1224 | 0.0005 | 0.0014* |
| MYH9 | IRefIndex_2015 | 44 | 6 | 0.1602 | 0.2033 | 0.6007 | 0.6447 |
| TP53 | CPDB | 975 | 163 | 0.2977 | 0.1728 | 0.0001 | 0.0004* |
| TP53 | STRING | 494 | 111 | 0.4338 | 0.2337 | 0.0001 | 0.0004* |
| TP53 | MultiNet | 346 | 62 | 0.3289 | 0.1922 | 0.0001 | 0.0004* |
| TP53 | PCNet | 2946 | 400 | 0.2311 | 0.1380 | 0.0001 | 0.0004* |
| TP53 | IRefIndex | 839 | 129 | 0.2953 | 0.1603 | 0.0001 | 0.0004* |
| TP53 | IRefIndex_2015 | 402 | 75 | 0.3153 | 0.1975 | 0.0001 | 0.0004* |
| PCDH10 | CPDB | 7 | 1 | 0.2500 | 0.3685 | 0.5672 | 0.6240 |
| PCDH10 | STRING | 0 | 0 | NA | NA | NA | NA |
| PCDH10 | MultiNet | 1 | 1 | NA | NA | NA | NA |
| PCDH10 | PCNet | 71 | 14 | 0.5212 | 0.2424 | 0.0017 | 0.0036* |
| PCDH10 | IRefIndex | 9 | 1 | 0.2500 | 0.3165 | 0.4456 | 0.5298 |
| PCDH10 | IRefIndex_2015 | 0 | 0 | NA | NA | NA | NA |
| ACTN2 | CPDB | 141 | 10 | 0.1000 | 0.1010 | 0.3685 | 0.4622 |
| ACTN2 | STRING | 143 | 16 | 0.2809 | 0.1407 | 0.0078 | 0.0137* |
| ACTN2 | MultiNet | 21 | 0 | NA | NA | NA | NA |
| ACTN2 | PCNet | 706 | 70 | 0.1741 | 0.1069 | 0.0007 | 0.0018* |
| ACTN2 | IRefIndex | 145 | 12 | 0.1398 | 0.1115 | 0.1907 | 0.2468 |
| ACTN2 | IRefIndex_2015 | 68 | 7 | 0.1526 | 0.1544 | 0.3782 | 0.4622 |
| FBN1 | CPDB | 32 | 4 | 0.1396 | 0.2116 | 0.6896 | 0.7057 |
| FBN1 | STRING | 117 | 9 | 0.0902 | 0.1109 | 0.5614 | 0.6240 |
| FBN1 | MultiNet | 6 | 2 | 0.4167 | 0.5283 | 0.6694 | 0.7013 |
| FBN1 | PCNet | 592 | 61 | 0.1455 | 0.1121 | 0.0382 | 0.0542 |
| FBN1 | IRefIndex | 67 | 7 | 0.1287 | 0.1550 | 0.5384 | 0.6235 |
| FBN1 | IRefIndex_2015 | 31 | 5 | 0.1626 | 0.2469 | 0.7689 | 0.7689 |
| PRKDC | CPDB | 283 | 56 | 0.3659 | 0.2125 | 0.0002 | 0.0006* |
| PRKDC | STRING | 72 | 19 | 0.4467 | 0.3041 | 0.0255 | 0.0392* |
| PRKDC | MultiNet | 90 | 14 | 0.4459 | 0.1948 | 0.0010 | 0.0023* |
| PRKDC | PCNet | 1542 | 207 | 0.2478 | 0.1382 | 0.0001 | 0.0004* |
| PRKDC | IRefIndex | 221 | 43 | 0.3460 | 0.2129 | 0.0011 | 0.0024* |
| PRKDC | IRefIndex_2015 | 123 | 26 | 0.3593 | 0.2401 | 0.0157 | 0.0256* |

Table S19 shows significant results in 21 of 24 known-gene tests and nine of 20 evaluable candidate tests. PIK3CA and TP53 showed significance across all six PPIs; APC and MYH9 showed significance in five and four PPIs, respectively. Among candidates, PRKDC showed significance in all six networks, ACTN2 in STRING and PCNet, and PCDH10 in PCNet. FBN1 showed no significant results after correction; its PCNet result had $q = 0.0542$.

Together, the analyses demonstrate reproducible model explanations and statistical support for external STAD-associated genes in many evaluated neighborhoods. Stability and biological relevance are distinct: FBN1 had stable explanations without significant neighbor enrichment. Neither the neighbor tests nor the high candidate prediction scores establish causal mechanisms or confirm the candidates as cancer drivers.

## 2.7 Sensitivity to negative labels and unlabeled candidates

We performed two separate analyses to assess sensitivity to uncertain negative labels and cancer-associated unlabeled competitors. The first changes the negative set and retrains NAMAT; the second retains the original trained models and changes only the ranking candidate pool.

### *2.7.1 Cancer-specific filtering of negative genes*

The original benchmark uses a shared set of 2,187 negative genes. For each cancer type, we removed negative genes associated with that cancer in either Open Targets (overall association score $\geq 0.20$) or curated DisGeNET (score $\geq 0.40$). Filtering used the union of both resources, so overlapping genes were removed only once. Open Targets release 26.09 was used except for BRCA, which used release 25.06 and the DisGeNET export of August 2025. Table S20 reports the filtering counts.

Table S20. Cancer-specific filtering of the original 2,187 negative genes. Open Targets (OT) uses association scores $\geq 0.20$ and DisGeNET (DG) uses curated association scores $\geq 0.40$. OT removed and DG removed are counts within the original negative set. Union removed counts each excluded gene once. BRCA uses Open Targets release 25.06; the other cancer types use release 26.09.

| Cancer | OT removed | DG Removed | Union Removed | Remaining |
|---|---|---|---|---|
| **BRCA** | 157 | 55 | 203 | 1984 |
| **BLCA** | 5 | 7 | 12 | 2175 |
| **LUAD** | 7 | 6 | 13 | 2174 |
| **LIHC** | 7 | 42 | 48 | 2139 |
| **THCA** | 0 | 1 | 1 | 2186 |
| **LUSC** | 5 | 1 | 6 | 2181 |
| **ESCA** | 6 | 0 | 6 | 2181 |
| **PRAD** | 2 | 29 | 31 | 2156 |
| **STAD** | 4 | 0 | 4 | 2183 |
| **COAD** | 1 | 1 | 2 | 2185 |
| **UCEC** | 2 | 0 | 2 | 2185 |
| **CESC** | 48 | 0 | 48 | 2139 |

NAMAT was retrained using each cancer-specific filtered negative set and the original positive genes, retaining the model configuration and five-fold cross-validation repeated over 10 runs. Each run-level AUPRC was averaged over five folds. Because filtering changes the labeled gene sets and the resulting splits, the original and filtered runs were treated as unpaired for the per-cancer comparisons. Two-sided Welch $t$ tests were calculated from the run-level means and standard deviations. The overall comparison used a two-sided Wilcoxon signed-rank test across the 12 cancer-specific mean AUPRC values, paired by cancer type.

Average AUPRC changed from 0.683 to 0.687, with no significant overall difference across cancer types ($p = 0.371$). BRCA, LIHC, and PRAD showed nominal per-cancer improvements. Most other changes were small, although LUSC decreased from 0.616 to 0.580 without reaching nominal significance. These findings indicate limited aggregate sensitivity to this particular filtering procedure, while allowing cancer-specific effects. They do not establish that the remaining negatives are free of unknown drivers. Moreover, because the labeled evaluation sets also change, this analysis does not isolate the effect of training-label contamination alone.

Table S21. AUPRC of NAMAT with the original and cancer-specific filtered negative sets. Values are mean ± SD over 10 runs, each averaged over five folds. Per-cancer $p$-values are from two-sided Welch $t$ tests; the overall $p$-value is from a two-sided Wilcoxon signed-rank test across cancer types. Asterisks indicate nominal per-cancer $p < 0.05$, without adjustment for multiple comparisons. Bold identifies the higher mean.

| Cancer | Original Negatives | Filtered Negatives | $p$ |
|---|---|---|---|
| **BRCA** | $0.802 \pm 0.006$ | $\mathbf{0.842} \pm 0.004$ | $< 0.001^*$ |
| **BLCA** | $0.789 \pm 0.005$ | $\mathbf{0.794} \pm 0.007$ | 0.110 |
| **LUAD** | $\mathbf{0.744} \pm 0.008$ | $0.743 \pm 0.007$ | 0.736 |
| **LIHC** | $0.595 \pm 0.019$ | $\mathbf{0.626} \pm 0.018$ | $0.002^*$ |
| **THCA** | $0.426 \pm 0.026$ | $\mathbf{0.430} \pm 0.024$ | 0.765 |
| **LUSC** | $\mathbf{0.616} \pm 0.053$ | $0.580 \pm 0.044$ | 0.137 |
| **ESCA** | $0.611 \pm 0.015$ | $\mathbf{0.612} \pm 0.016$ | 0.863 |
| **PRAD** | $0.827 \pm 0.009$ | $\mathbf{0.836} \pm 0.009$ | $0.048^*$ |
| **STAD** | $\mathbf{0.779} \pm 0.009$ | $0.773 \pm 0.009$ | 0.179 |
| **COAD** | $\mathbf{0.610} \pm 0.008$ | $0.606 \pm 0.006$ | 0.261 |
| **UCEC** | $\mathbf{0.743} \pm 0.014$ | $0.740 \pm 0.020$ | 0.687 |
| **CESC** | $0.655 \pm 0.030$ | $\mathbf{0.666} \pm 0.043$ | 0.551 |
| **Average** | 0.683 | **0.687** | 0.371 |

### 2.7.2 *Filtering cancer-associated unlabeled competitors*

To assess candidate-pool sensitivity separately, we retained the original NAMAT models and prediction scores. For each cancer type, unlabeled genes were excluded from the ranking pool if they were listed in NCG 6.1, COSMIC Cancer Gene Census v104, KEGG cancer pathways, cancer-related OMIM entries, Open Targets associations for that cancer (score $\geq 0.20$), or curated DisGeNET associations for that cancer (score $\geq 0.40$). Held-out positive test genes were never removed. Thus, the comparison changes only the unlabeled competitors and requires no retraining.

Hits@$K$ was recalculated for $K \in \{1, 3, 5, 10, 20, 50\}$ using the original and filtered candidate pools. Results were averaged over five folds within each of 10 runs and then across runs. AUPRC on the original labeled test sets was unaffected by this candidate-pool filtering.

Table S22 shows higher held-out-positive recovery after filtering for every cancer type and evaluated cutoff. Average Hits@1, Hits@3, Hits@5, Hits@10, Hits@20, and Hits@50 changed from 0.41, 1.06, 1.44, 2.20, 2.87, and 4.23 to 0.84, 1.96, 2.71, 3.88, 5.32, and 7.18, respectively. Approximately 2,910 unlabeled competitors were removed per cancer on average.

Removing unlabeled competitors while retaining every held-out positive cannot decrease Hits@$K$. The increases therefore quantify sensitivity to candidate-pool composition rather than improved model predictions or independent confirmation of the excluded genes as drivers. The original open-world ranking remains the primary evaluation; the filtered analysis illustrates how cancer-associated unlabeled genes can occupy high-ranking positions without receiving credit as recovered positives.

Table S22. Hits@$K$ with the original and filtered candidate pools, using unchanged NAMAT models and prediction scores. O denotes the original pool and F the pool after removing cancer-associated unlabeled competitors. Held-out positive genes were never removed. Values are means over 10 runs, each averaged over five folds. Removed is the number of excluded unlabeled genes; the average removal count is rounded to the nearest gene.

| Cancer | Pool | Hits@1 | Hits@3 | Hits@5 | Hits@10 | Hits@20 | Hits@50 | Removed |
|---|---|---|---|---|---|---|---|---|
| **BRCA** | O | 0.46 | 1.08 | 1.86 | 3.66 | 4.52 | 8.10 | – |
| | F | 1.00 | 2.94 | 4.72 | 7.82 | 12.80 | 18.10 | 4100 |
| **BLCA** | O | 0.36 | 1.74 | 2.70 | 4.44 | 5.90 | 7.30 | – |
| | F | 1.00 | 2.94 | 4.54 | 6.52 | 7.94 | 10.20 | 2745 |
| **LUAD** | O | 0.28 | 0.72 | 1.00 | 1.48 | 2.28 | 4.78 | – |
| | F | 0.84 | 2.14 | 3.02 | 4.44 | 7.00 | 10.32 | 2758 |
| **LIHC** | O | 0.26 | 0.80 | 0.96 | 1.18 | 1.72 | 3.52 | – |
| | F | 0.74 | 1.24 | 1.54 | 3.10 | 4.20 | 5.32 | 3018 |
| **THCA** | O | 0.40 | 0.68 | 0.80 | 1.04 | 1.26 | 1.60 | – |
| | F | 0.64 | 1.16 | 1.32 | 1.78 | 2.42 | 2.84 | 2717 |
| **LUSC** | O | 0.20 | 0.36 | 0.44 | 0.56 | 0.70 | 1.00 | – |
| | F | 0.54 | 0.82 | 0.94 | 1.04 | 1.28 | 1.86 | 2780 |
| **ESCA** | O | 0.86 | 1.60 | 2.08 | 2.72 | 3.36 | 3.94 | – |
| | F | 0.88 | 2.12 | 2.86 | 3.56 | 4.48 | 5.70 | 2729 |
| **PRAD** | O | 0.28 | 0.82 | 1.06 | 2.12 | 2.70 | 4.26 | – |
| | F | 0.84 | 1.88 | 2.60 | 3.38 | 5.20 | 6.78 | 2982 |
| **STAD** | O | 0.28 | 1.26 | 1.46 | 2.14 | 3.24 | 4.82 | – |
| | F | 0.96 | 2.40 | 3.22 | 4.52 | 5.52 | 7.38 | 2735 |
| **COAD** | O | 0.94 | 2.14 | 2.58 | 3.22 | 4.12 | 5.38 | – |
| | F | 1.00 | 2.42 | 3.34 | 4.38 | 5.76 | 8.34 | 2636 |
| **UCEC** | O | 0.44 | 1.22 | 1.68 | 2.76 | 3.38 | 4.76 | – |
| | F | 0.92 | 2.24 | 3.12 | 4.58 | 5.72 | 7.64 | 2725 |
| **CESC** | O | 0.20 | 0.26 | 0.68 | 1.10 | 1.20 | 1.34 | – |
| | F | 0.74 | 1.22 | 1.34 | 1.38 | 1.46 | 1.70 | 2996 |
| **Average** | O | 0.41 | 1.06 | 1.44 | 2.20 | 2.87 | 4.23 | – |
| | F | 0.84 | 1.96 | 2.71 | 3.88 | 5.32 | 7.18 | 2910 |

## 2.8 Computational-cost benchmark

Computational costs were measured in Google Colab on an NVIDIA L4 GPU using Python 3.13.16, PyTorch 2.11.0, CUDA 13.0, and PyTorch Geometric 2.8.1. All methods used one fixed STAD split (1,536 training, 272 validation, and 452 test genes), five model seeds (42–46), and a common universe of 13,627 genes. Hyperparameters were fixed, with a common maximum of 380 epochs and early-stopping patience of 80 based on validation loss. Training time includes validation but excludes data preparation and hyperparameter search. GPU synchronization was used around timing. Inference used the best-validation model in evaluation mode without gradients, averaging 100 full-graph passes after 10 warm-up passes. Peak training memory denotes allocated GPU tensor memory, including resident inputs. Baselines were PyTorch reimplementations.

Table S23. Computational cost on STAD using an NVIDIA L4 GPU. Times are mean ± SD over five seeds. Inference scores all 13,627 genes. Training memory is rounded to whole MB; SD was zero at this precision. Limit indicates runs reaching 380 epochs. MTGCN additionally required 621.4 s for one-time learned DeepWalk preprocessing, excluded from training time.

| **Method** | **Training** (s) | **Time/epoch** (ms) | **Inference** (ms) | **Training Memory** (MB) | **Limit** |
|---|---|---|---|---|---|
| **NAMAT** | 24.0 ± 5.8 | 82.9 ± 0.7 | 12.74 ± 0.13 | 733 | 0/5 |
| **Uniform** | 21.2 ± 1.8 | 79.6 ± 0.6 | 11.84 ± 0.09 | 712 | 0/5 |
| **Global** | 21.6 ± 2.7 | 74.7 ± 0.5 | 10.50 ± 0.09 | 672 | 0/5 |
| **PPI Concat** | 27.2 ± 4.2 | 84.7 ± 0.4 | 13.30 ± 0.09 | 824 | 1/5 |
| **Concat Attention** | 28.7 ± 5.0 | 84.5 ± 0.7 | 13.49 ± 0.10 | 825 | 0/5 |
| **NAMAT–PCNet** | 15.6 ± 4.5 | 55.0 ± 0.3 | 6.22 ± 0.05 | 607 | 0/5 |
| **PCNet without Gate** | 17.1 ± 2.3 | 50.9 ± 0.1 | 5.54 ± 0.02 | 593 | 2/5 |
| **EMOGI** | 2.7 ± 0.0 | 7.1 ± 0.1 | 0.80 ± 0.01 | 96 | 5/5 |
| **MTGCN** | 197.8 ± 0.1 | 520.5 ± 0.2 | 11.54 ± 0.03 | 670 | 5/5 |
| **EMGNN** | 59.3 ± 13.8 | 303.3 ± 0.6 | 104.80 ± 0.17 | 3735 | 0/5 |
| **MODIG** | 139.8 ± 0.2 | 368.0 ± 0.3 | 93.13 ± 0.04 | 8042 | 5/5 |
| **DGHNN** | 160.4 ± 19.4 | 445.1 ± 2.1 | 615.95 ± 0.54 | 1783 | 4/5 |

NAMAT added approximately 4% time per epoch, 8% inference time, and 3% training memory relative to Uniform. Compared with Concat Attention, NAMAT required approximately 2% less time per epoch, 6% less inference time, and 11% less training memory. Compared with EMGNN, it required approximately 27% of the time per epoch, 12% of the inference time, and 20% of the training memory. Several baselines reached the common epoch limit, so their total times represent the benchmark budget rather than training to convergence under their original schedules. MODIG includes 175 additional isolated nodes to match the shared gene universe. These results characterize the stated implementations on one STAD split and GPU type.

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